

EEBus UC Technical Specification

Power Demand Forecast

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1 Scope of the document

This document describes the Use Case "Power Demand Forecast" (short-name: PODF). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonizing energy demand and availability.

DSO

Abbreviation for "Distribution System Operator"

Energy Guard

A logical or physical unit capable of managing the energy demand of connected appliances.

ESP

Abbreviation for "Energy Service Provider"

EV

Abbreviation for "Electric Vehicle"

PODF

Power Demand Forecast (short name of this Use Case)

POEN

Short name of the Use Case "Power Envelope"

Scenario

Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization

Reusable data collection for a specific functionality.

SPINE

Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

TOUT

Short name of the Use Case "Time of Use Tariff"

UC

Abbreviation for "Use Case"

1.3 Requirements**1.3.1 Requirements wording**

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[PODF-xyz]

e.g.: [PODF-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

The electricity grid is more and more challenged by energy consuming devices such as electric vehicles (EVs), heat pumps, etc. on the one hand and transition to regenerative energy production on the other hand. To enable this, energy demand and energy production needs to be harmonized. It is essential for the grid-side to have information about the power consumption or production of buildings. The current power consumption / production on building level can be retrieved via the Use Case "Monitoring of Grid Connection Point", but knowing a predicted energy behaviour of the buildings can improve the overall load management of the grid-side, especially for the DSO. This Use Case offers a power over time forecast that may be sent to the relevant Actors Energy Broker and Transmission Broker (as the functional Actor for the DSO) by a customer energy manager (CEM).

The reference point for the power demand forecast is the building's grid connection point.

Additionally, the Actor CEM can provide the maximum power consumption and production that the building is allowed to consume / produce due to the customer's contract.

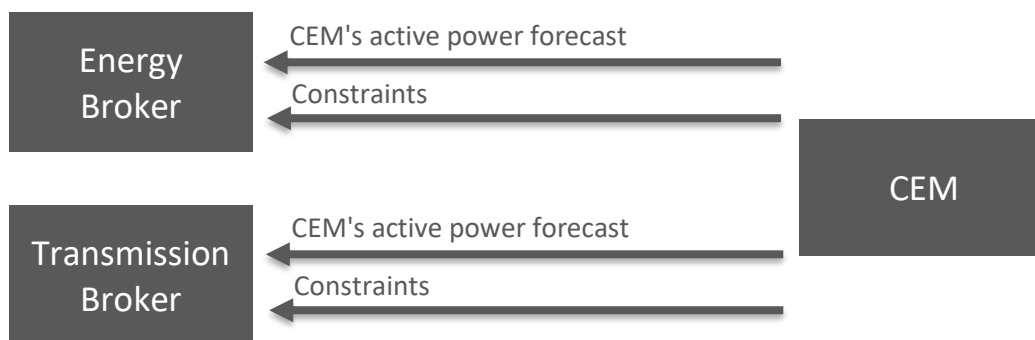


Figure 1: High-Level Use Case functionality overview

2.2 Detailed background information and rules

2.2.1 Information and rules regarding the Use Case's Actors

As the previous section introduces, there are three Actors within this Use Case (Energy Broker, Transmission Broker and CEM). The Actor CEM is always needed for the Use Case's functionality. But whether Energy Broker, Transmission Broker or both are part of the UC depends on the overall electricity grid setup, the country's regulations and the partners the customer chose for the electricity supply. The CEM SHALL be able to send its active power forecast to both Actors.

2.2.2 Structure of a power forecast

The Actor CEM can forecast the anticipated power consumption and production of the building with a power forecast curve. The power forecast describes power values over time. The time is split into slots ([PODF-003]). For each slot at least the expected (average) power consumption/production P_{exp} ([PODF-004]) SHALL be given. The minimum power consumption/production P_{min} ([PODF-005]) or maximum power consumption/production P_{max} ([PODF-006]) MAY be given. Figure 2 shows an example.

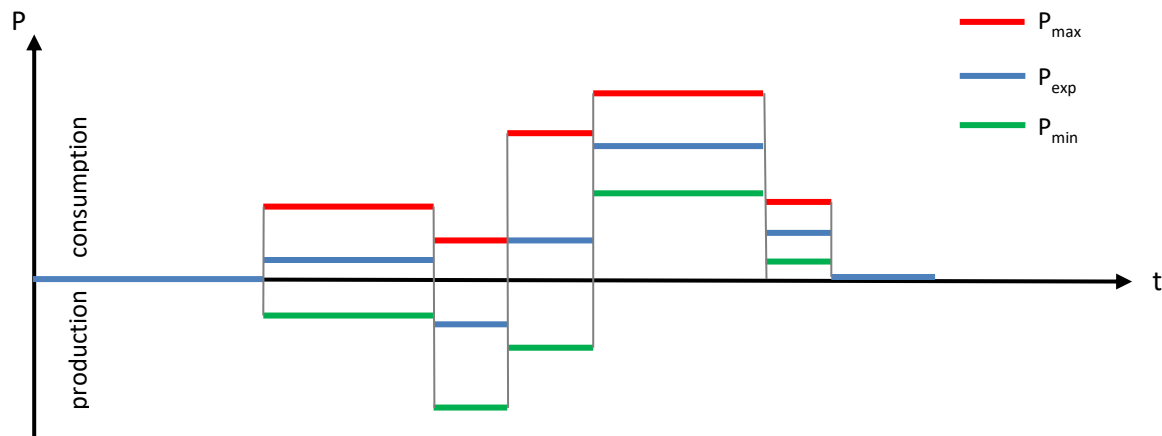


Figure 2: Power forecast example: power (P) curves over time (t)

The values of the power forecast curve include the typical base load of the building.

The values $P_{\min}$ and $P_{\max}$ are used to model an uncertainty factor of the forecast curve.

The building's power consumption/production for every time t SHOULD NOT be below the corresponding value of $P_{\min}(t)$ ([PODF-010/1]) and SHOULD NOT be above $P_{\max}(t)$ ([PODF-010/2]).

2.2.3 Considered time horizon of power forecast

The time horizon of the CEM's active power forecast curve SHALL always cover at least the next 6 hours. Hence, latest 6 hours before the last sent curve's time ends, an update SHALL be provided by the CEM.

The time period of a power forecast curve SHALL start at the current time*¹ ("now") and end latest in 48 hours.

*¹: The following rules SHALL apply for the first slot of the power forecast curve:

- start time $\leq$ "now"
- start time + duration $>$ "now"

Note: In case of inaccurate time synchronization, a receiving appliance could receive a start time of the first slot that lies in the future, or an end time (start time + duration) of the first slot that lies in the past (both are not allowed according to the rules above). However, the receiving device should calculate with a certain time tolerance in order not to reject an actually correct power forecast curve as incorrect due to the inaccurate time synchronization.

Note: As there must always be a power forecast curve available that covers the next 6 hours, an update of the power forecast curve SHOULD cover at least the next 12 hours in order to not be obliged to send updates too often. Example: If an 8 hour forecast curve is sent, the next update must be sent latest after 2 hours. If a 12 hour forecast curve is sent, the next update must be sent latest after 6 hours.

The start time of each slot SHALL be expressed as UTC value ([PODF-001]), whereas the slot's duration SHALL be presented as relative time ([PODF-002]). A power forecast SHALL NOT have time gaps between two slots ([PODF-007]).

Note: All times are transmitted as UTC time. Hence, local times must be derived from that by the Actors themselves, if needed.

Note: A power forecast curve that covers a long time period may have a shorter time resolution (i.e. more slots) for the near future, while slots of the distant future have a coarser time resolution. In general, the time resolution should be chosen to express significant changes of the power.

Note: It can be expected that predicted power values of the distant future are less reliable in comparison to those of the near future in many cases.

2.2.4 Updating the power forecast

The CEM will operate the devices (incl. baseload considered) according to the power demand forecast. Nevertheless, if the power demand differs significantly in a time period of at least 15 minutes, the CEM SHOULD send a new power demand forecast (with the time horizon requirements from section 2.2.3 fulfilled).

Note: Changes that are in the near future should be more likely to result in an update than those that are in the more distant future.

Note: A significant change may be 10kW for the residential case.

Note: To make sure the CEM implementation works in different residential setups (x kW), or in commercial setups, this value should be considered as configurable (e.g., in the CEM's user interface).

As described in section 2.2.3, at latest 6 hours before the last sent curve ends, a new power forecast SHALL be sent to all interested Actors.

There SHALL be at least one update of the power forecast curve within 24 hours.

Note: Requesting the CEM's current power forecast curve is possible at any time.

2.3 Actors

2.3.1 Energy Broker

The Actor Energy Broker is located at the contract partner of the customer. Within the Use Case "Time of Use Tariff" the Energy Broker sends incentive tables to the Actor CEM which may trigger the CEM to re-calculate its overall power consumption plan. This power demand forecast is then sent to the Actor Energy Broker for information purposes.

2.3.2 Transmission Broker

The Actor Transmission Broker may send incentive tables with the Use Case "Time of Use Tariff" to the Actor CEM as well (containing only the transmission fee). This may trigger the CEM to re-calculate its overall power consumption plan. This power demand forecast is then sent to the Actor Transmission Broker. The Transmission Broker checks whether this prediction plan fits into its overall electricity grid management plan. If not, this could lead to a power limitation of the building by sending a power limit curve to the CEM using the Use Case "Power Envelope".

2.3.3 CEM

The Actor CEM manages the devices in the building and calculates a power demand forecast of the known appliances as well as the anticipated baseload of the building.

2.4 Scenarios

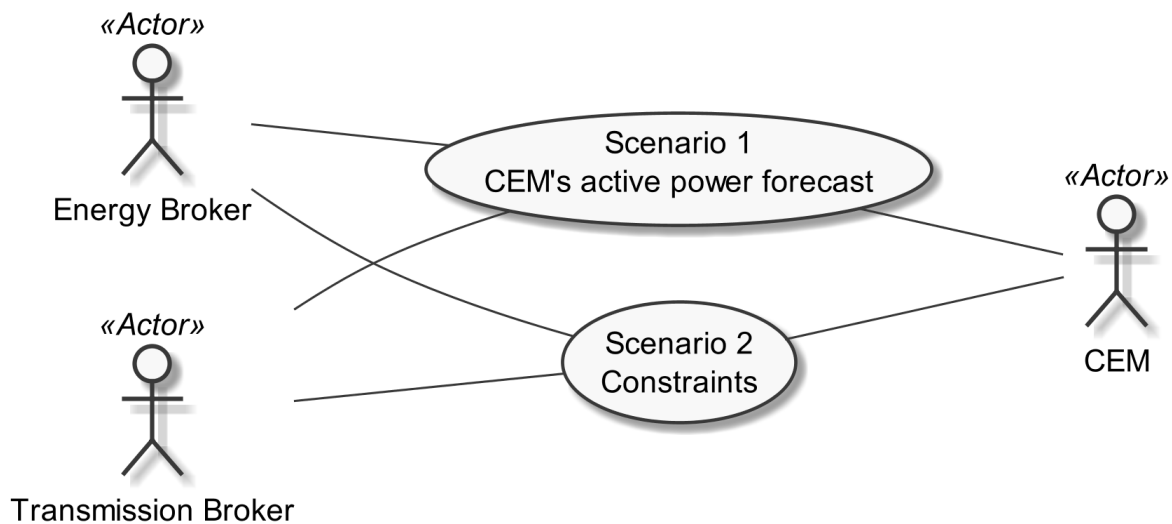


Figure 3: Scenario overview

Scenario Number	Scenario Name	Energy Broker	Transmission Broker	CEM
1	CEM's active power forecast	M	M	M
2	Constraints	R	R	R

Table 1: Scenario implementation requirements for Actors

2.4.1 Scenario 1 - CEM's active power forecast

2.4.1.1 Description

The CEM calculates the power consumption and production of the building based on the information from the connected devices, learning algorithms about typical base load, etc.

As described in section 2.2.2 the Active Power Forecast

- SHALL contain the expected (average) power consumption/production ([PODF-004])
- MAY contain the minimum power consumption/production $P_{\min}$ ([PODF-005])
- MAY contain the maximum power consumption/production $P_{\max}$ ([PODF-006]).

This power forecast is updated regularly by the CEM and can be requested by clients interested in this information as well as it is sent automatically to subscribed clients.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Active Power Forecast	A power over time curve of the probable power consumption and production of the building's appliances managed by the CEM.	M	[PODF-011]

Table 2: Scenario 1 - CEM's active power forecast - Data point list

2.4.1.2 Conditions

Triggering Event:

The Actor CEM calculated a new power forecast.

Pre-condition:

The CEM's power forecast is outdated.

Post-condition:

The Actor CEM sent its updated power forecast to all subscribed clients.

2.4.2 Scenario 2 - Constraints

2.4.2.1 Description

The constraints depicted in this Scenario inform clients about the maximum power that the building is allowed to consume and produce due to the customer's contract.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Contractual Consumption Nominal Max	The nominal maximum active (real) power the building is allowed to consume due to the customer's contract.	R	[PODF-021]
Contractual Production Nominal Max	The nominal maximum active (real) power the building is allowed to produce due to the customer's contract.	R	[PODF-022]

Table 3: Scenario 2 - Constraints - Data point list

Note: If this Scenario is supported, at least one of the values stated above SHALL be supported.

2.4.2.2 Conditions

Triggering Event:

The Actor Energy Broker or Transmission Broker is interested in the constraints of the Actor CEM.

Pre-condition:

The Actor Energy Broker or Transmission Broker does not know the constraints of the Actor CEM.

Post-condition:

The Actor Energy Broker or Transmission Broker knows the constraints of the Actor CEM.

2.5 Dependencies to other Use Cases

2.5.1 "Limitation of Power Consumption"

The Use Case "Limitation of Power Consumption" is used to limit the power consumption of an appliance or a building*¹ (by the use of a CEM).

*¹: Please consider country specific regulations regarding the application scope of the Use Case LPC.

If the Actor CEM supports the Use Case "Limitation of Power Consumption", and the data point "Contractual Consumption Nominal Max" is supported in this Use Case as well as in the Use Case "Limitation of Power Consumption", the content of [LPC-042] and [PODF-021] SHALL be the same.

The Actor Transmission Broker MAY support the Use Case "Limitation of Power Consumption".

The Actor CEM MAY support the Use Case "Limitation of Power Consumption".

The Actor Transmission Broker of this Use Case acts as Actor Energy Guard within the Use Case "Limitation of Power Consumption".

The Actor CEM of this Use Case acts as Actor Controllable System within the Use Case "Limitation of Power Consumption".

2.5.2 "Limitation of Power Production"

The Use Case "Limitation of Power Production" is used to limit the power production of an appliance or a building*¹ (by the use of a CEM).

*¹: Please consider country specific regulations regarding the application scope of the Use Case LPP.

If the Actor CEM supports the Use Case "Limitation of Power Production", and the data point "Contractual Production Nominal Max" is supported in this Use Case as well as in the Use Case "Limitation of Power Production", the content of [LPP-042] and [PODF-022] SHALL be the same.

The Actor Transmission Broker MAY support the Use Case "Limitation of Power Production".

The Actor CEM MAY support the Use Case "Limitation of Power Production".

The Actor Transmission Broker of this Use Case acts as Actor Energy Guard within the Use Case "Limitation of Power Production".

346 The Actor CEM of this Use Case acts as Actor Controllable System within the Use Case "Limitation of
347 Power Production".

348

349 **2.5.3 "Monitoring of Grid Connection Point"**

350 This dependency only holds valid for the Actors Energy Broker and Transmission Broker and is only
351 for informative purposes.

352 The Use Case "Monitoring of Grid Connection Point" enables the Actors Energy Broker and
353 Transmission Broker to monitor the power consumption or production at the Grid Connection Point.

354 The Actor Energy Broker MAY support the Use Case "Monitoring of Grid Connection Point".

355 The Actor Energy Broker of this Use Case acts as Actor Monitoring Appliance within the Use Case
356 "Monitoring of Grid Connection Point".

357 The Actor Transmission Broker MAY support the Use Case "Monitoring of Grid Connection Point".

358 The Actor Transmission Broker of this Use Case acts as Actor Monitoring Appliance within the Use
359 Case "Monitoring of Grid Connection Point".

360

361 **2.5.4 "Power Envelope"**

362 The Use Case "Power Envelope" enables the Actor Transmission Broker to limit the building's power
363 consumption or production via the Actor CEM with power-over-time curves. The power envelope can
364 include up to four different limit curves:

- 365 - Minimum power consumption
- 366 - Maximum power consumption
- 367 - Minimum power production
- 368 - Maximum power production

369 The Actor Transmission Broker SHOULD support the Use Case "Power Envelope".

370 The Actor CEM SHOULD support the Use Case "Power Envelope".

371

372 **2.5.5 "Time of Use Tariff"**

373 The Use Case "Time of Use Tariff" is used by the Actors Energy Broker and Transmission Broker to
374 inform the Actor CEM about the upcoming incentive (e.g. price) values for the building's power
375 consumption or production.

376 The Actor Energy Broker SHOULD support the Use Case "Time of Use Tariff".

377 The Actor Transmission Broker MAY support the Use Case "Time of Use Tariff".

378 The Actor CEM SHOULD support the Use Case "Time of Use Tariff".

379

2.6 Further information and rules

2.6.1 Sign conventions

The following values of this Use Case follow the passive sign convention (positive values indicate consumption, negative values indicate production) ([PODF-008]):

- Active Power Forecast ([PODF-011])

The following values are always positive. Their name indicates whether they are used for consumption or production direction ([PODF-009]):

- Contractual Consumption Nominal Max ([PODF-021]): consumption direction
- Contractual Production Nominal Max ([PODF-022]): production direction

2.7 Assumptions and Prerequisites

None.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.3.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.3.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "powerDemandForecast". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
--------------	---------	------------------------------

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 4: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

465 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 466 - M, O
- 467 - M \W, O \W

468 The description (rules) of the Element details in such a case the presence requirement. A typical
469 example is a list-based Element where the "first" (or "last") Element has a different presence
470 requirement than the other list Elements.

471

472 **3.1.3.3 Presence indications for "Content of Function..." tables**

473 This section only defines rules for the server side.

474 Elements that are marked with "M" SHALL be supported by the server in case of readable as well as
475 writeable data. In case of writeable data (marked with "M \W") the server does not need to set the
476 Element, because the Element is set only by the client.

477 The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- 478 - "R" means that the data SHOULD be provided by the server.
- 479 - "O" means that the data MAY be provided by the server.
- 480 - "F" means that the data SHALL NOT be provided by the server.

481 The following applies for writeable data that is exchanged in a "write" operation:

- 482 - "R" means that the data SHOULD be supported. In other words: If the client writes the
483 Element, the server SHOULD accept those messages and the contained Elements.
- 484 - "O" means that the data MAY be supported. In other words: If the client writes the Element,
485 the server MAY accept those messages and the contained Elements.

486 The following applies for Elements that are not listed in the Actor section:

- 487 - In case of a received "read" request: The according Element MAY be set in the reply.
- 488 - In case of a received "write" operation: The server MAY ignore the Element.
- 489 - In case of a "notify" operation to be created: The server MAY set the Element.

490 Note: The server will only accept write operations if the result fulfils the server Function
491 requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY
492 result in an error message.

493 M, R or O may be combined with the suffix "(event)" to express that a supported Element or value
494 only has to be supported during a certain event and hence does not need to be present at all times. If
495 the event is not active the Element may be omitted or another value may be set. In most cases a
496 High-Level requirement reference for the event is given in the rules column.

497 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 498 - M, O
- 499 - M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
No cardinality indication.
2. X (a..b)
This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..)
This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{...}]: M/R/O [W]\C]	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*#(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 5: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3

times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#{1..}>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

597 Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g.
 598 In this case, the uniqueness rule applies for "<p5>" likewise.

599 Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

600

601 3.1.3.6.3 Placeholder expressions with cardinalities

602 For some Identifiers, more than one placeholder is needed. Several notations are used for this
 603 purpose, which make use of cardinality expressions. The general notation is as follows:

604 1. <xM#(a..b)>

605 This is equivalent to this explicit definition:

606 At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

607 This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a"
 608 distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the
 609 respective Identifier.

610 Additionally, the following notations may occur:

611 2. <xM#(a..)>

612 This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

613 3. <xM#(..b)>

614 This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

615 "<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the
 616 other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to
 617 "b" distinct values.

618 Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*(a..b)> etc.

619 Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT
 620 mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers
 621 (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given,
 622 it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does
 623 NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a
 624 Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

625

626 3.1.3.6.4 References to placeholders and relations

627 According to the styles for placeholders that can be referenced, an enumeration value "e" can refer
 628 to a particular placeholder:

629 1. e(-><xM>)

630 2. e(-><xM#S>)

631 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)

"b"(-><m2#2>)

"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>

4. <yN->xM#S>

5. <yN#T->xM>

6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 6: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
 - o within the tables in the "Server data - Resources" sections:
 - the server SHALL support at least one of the listed values.
 - o within the tables in the "Client data - Specializations" sections:
 - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
 - o within the tables in the "Server data - Resources" sections:
 - the server SHALL support all listed values.
 - o within the tables in the "Client data - Specializations" sections:
 - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

696 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
 697 MAY deviate from this, e.g. "(1024)".

698

699 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 700 **Specialization..." tables**

701 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
 702 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
 703 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
 704 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
 705 background colour. This row contains the following parts:

706 <Feature Type>. <Function>.[<list entry instance name>.]

707 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 708 could be:

709 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
 710 deviceConfigurationKeyValueDescriptionData.

711 In the following rows, only the names of the Elements are stated, without the prefix described above.

712

713 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
 714 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement").
 715 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
 716 "Function "measurementListData"). These sections contain one table with all Elements needed for
 717 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
 718 colour. This row contains the following parts:

719 <Feature Type>. <Function>.[<list entry instance name>.]

720 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 721 could be:

722 Measurement. measurementDescriptionListData. measurementDescriptionData.

723 In the following rows, only the names of the Elements are stated, without the prefix described above.

724

725 For both kinds of tables, the following applies:

726 - Parent Elements are marked with a dot at the end of the name:

727 <parent Element>.

728 E.g.:

729 value.

730 - If there are sub-Elements, they are described in own rows with the name of the parent
 731 Element as prefix, separated by a dot and a blank space:

<parent Element>. <sub-Element>

E.g.:

value. number

3.1.4 Rules for "Feature Types and Functions..." tables

3.1.4.1 Presence indications for "Feature Types and Functions..." tables

The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 7: Presence indication of Feature Types and Functions support

If at least one Function of a Feature has the presence indication "M", it is mandatory to support the Feature.

3.1.4.2 Rules for "Possible operations" column

Within the "Feature Types and Functions..." tables the column "Possible operations" state whether the Function is read- or writeable (as defined in the detailed discovery mechanism, see [ProtocolSpecification]).

If the "partial" concept (also called "restricted function exchange") SHALL be supported, the following notation is used (separated for read and write access):

read (M). partial (M)

write (M). partial (M)

If the "partial" concept SHOULD be supported, the following notation is used:

read (M). partial (R)

write (M). partial (R)

If the "partial" concept MAY be supported, the following notation is used:

read (M). partial (O)

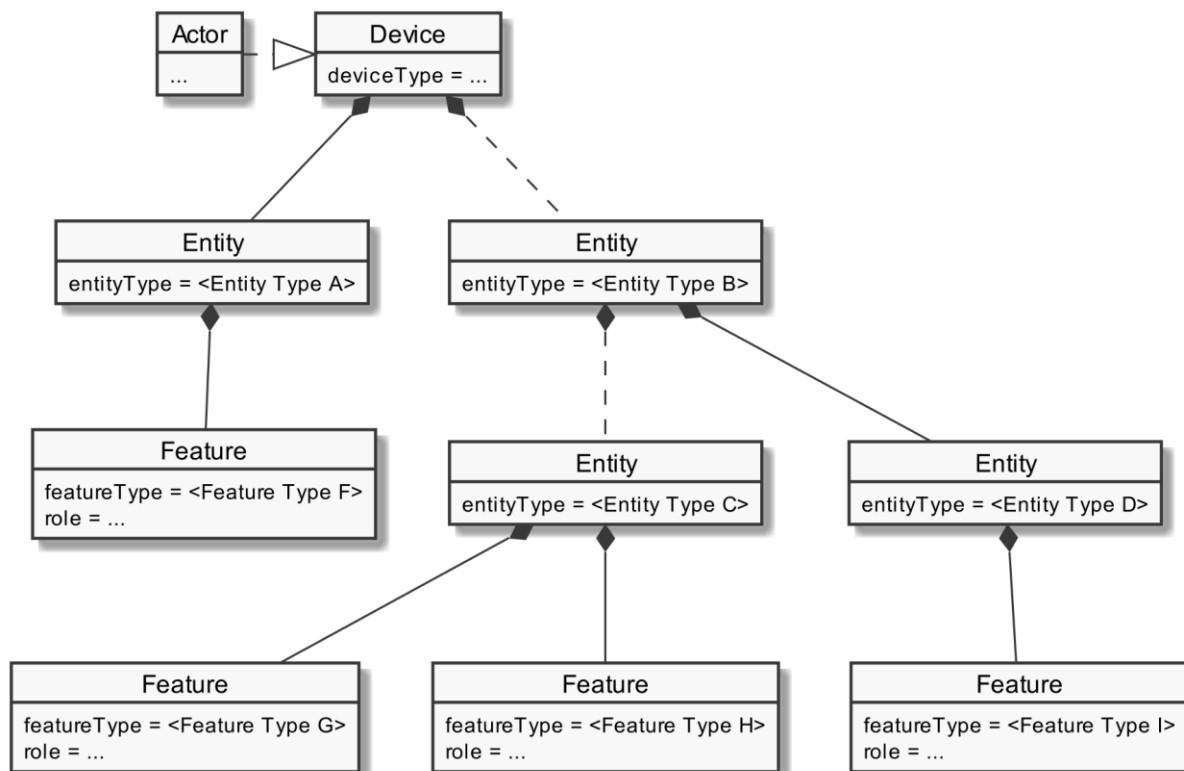
write (M). partial (O)

The server can decide whether a notification is submitted complete or partial (as described in [ProtocolSpecification]) if not defined differently within this Use Case Specification.

3.1.5 "Actor ... overview" diagram rules

Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

764 For the following Actor overview example, a brief description of the graphical symbols will be
 765 described.



766
 767 *Figure 4: Actor overview example*

768 The solid lines in the figure represent an immediate parent-childhood relation: The Entity with
 769 "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the
 770 Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

771 The dashed lines in the figure express that there MAY be additional Entities between the shown
 772 Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity
 773 with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between
 774 the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

775

776 3.1.6 Specializations

777 Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset
 778 necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data
 779 from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a
 780 Specialization defines a dataset that may encompass multiple related Functions from one or more
 781 different Features.

782 As different Use Cases sometimes share similar requirements, Specializations are also important
 783 from a re-usability perspective. This approach is used to improve consistency across Use Cases and
 784 avoid multiple variances of basically the same dataset. This is especially important in the case when
 785 an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two
 786 different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula:
$$\text{number} * 10^{\text{scale}}$$

3.1.8.2 Rules regarding the usage of time-related information

Timestamps used within this Use Case SHALL be presented as absolute times ([PODF-001]). This means relative times SHALL NOT be used for timestamps.

Durations used within this Use Case SHALL be presented as relative times ([PODF-002]).

Note: This means the Actors need to have a common understanding of absolute times. It is strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal Time" (UTC).

3.1.8.3 Applied sign convention

For the applied sign convention, please see section 2.6.1.

3.2 Actors

3.2.1 Energy Broker

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "EnergyBroker" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Energy Broker's resource hierarchy.

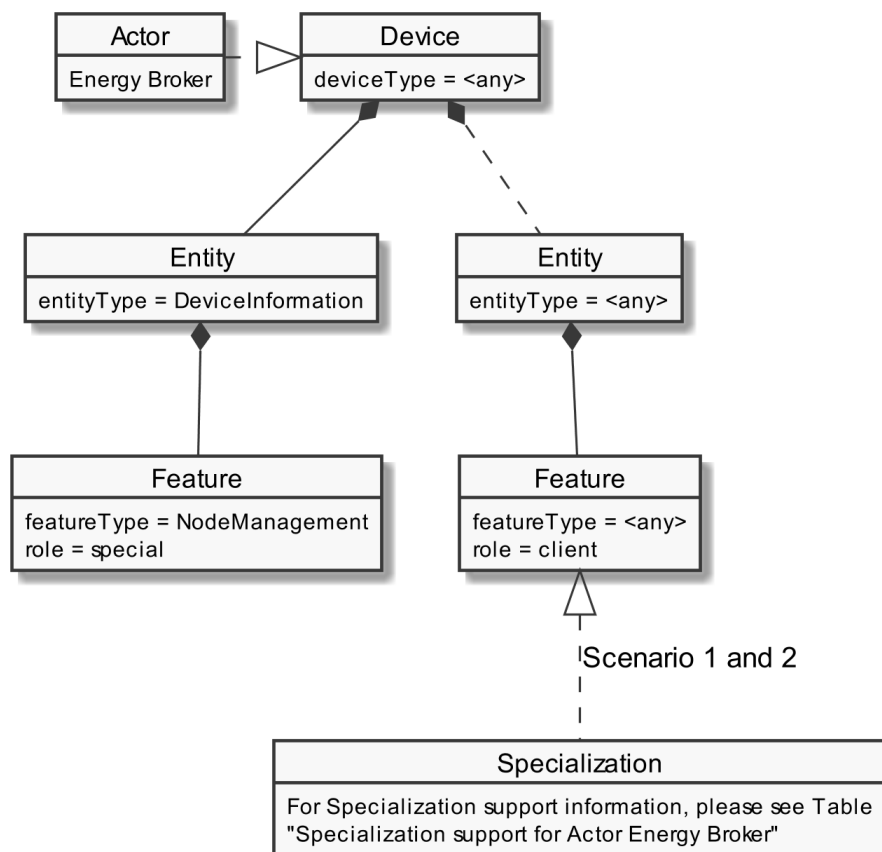


Figure 5: Actor "Energy Broker" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
TimeSeries_ActivePowerForecast	1	Table 9
ElectricalConnection_EntityContractualConsumptionNominalMax	2	Table 10
ElectricalConnection_EntityContractualProductionNominalMax	2	Table 11

Table 8: Specialization support for Actor Energy Broker

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "TimeSeries"

3.2.1.3.1.1 Specialization "TimeSeries_ActivePowerForecast"

Scenario [...] : M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries.timeSeriesDescriptionListData.timeSeriesDescriptionData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
1: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
1: O	measurementId	<m*#(1..1)>	FOREIGN IDENTIFIER.

			If measurementId is used, it SHALL link to a measurement value of the total active power.
1: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
1: M	scopeType	"activePowerForecast"	
1: M	TimeSeries. timeSeriesListData. timeSeriesData.		
1: M	timeSeriesId	<ts1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: F	timePeriod.		
1: M	timeSeriesSlot.		[PODF-003]
1: M	timeSeriesSlot. timeSeriesSlotId		
1: M	timeSeriesSlot. timePeriod.		
1: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[PODF-001]
1: M	timeSeriesSlot. duration		[PODF-002], [PODF-007] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
1: M	timeSeriesSlot. value.		[PODF-004], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal. See [Scaled number rules].
1: M	timeSeriesSlot. value. number		
1: M	timeSeriesSlot. value. scale		
1: M	timeSeriesSlot. minValue.		[PODF-005], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
1: M	timeSeriesSlot. minValue. number		

1: M	timeSeriesSlot. minValue. scale		
1: M	timeSeriesSlot. maxValue.		[PODF-006], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
1: M	timeSeriesSlot. maxValue. number		
1: M	timeSeriesSlot. maxValue. scale		

Table 9: Content of Specialization "TimeSeries_ActivePowerForecast" at Actor Energy Broker

3.2.1.3.2 Topic "ElectricalConnection"

3.2.1.3.2.1 Specialization "ElectricalConnection_EntityContractualConsumptionNominalMax"

Scenario {...}: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
2: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicId	<cc1#(1..1)>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicContext	"entity"	
2: M	characteristicType	"contractualConsumptionNominalMax"	
2: M	value.		[PODF-009], [PODF-021] [Scaled number rules]
2: M	value. number	≥0	SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	unit	"W"	

Table 10: Content of Specialization "ElectricalConnection_EntityContractualConsumptionNominalMax" at Actor Energy Broker

3.2.1.3.2.2 Specialization "ElectricalConnection_EntityContractualProductionNominalMax"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#{1..1}>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicId	<cc2#{1..1}>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicContext	"entity"	
2: M	characteristicType	"contractualProductionNominalMax"	
2: M	value.		[PODF-009], [PODF-022] [Scaled number rules]
2: M	value. number	≥0	SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	unit	"W"	

Table 11: Content of Specialization "ElectricalConnection_EntityContractualProductionNominalMax" at Actor Energy Broker

3.2.2 Transmission Broker

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "Transmission Broker" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Transmission Broker's resource hierarchy.

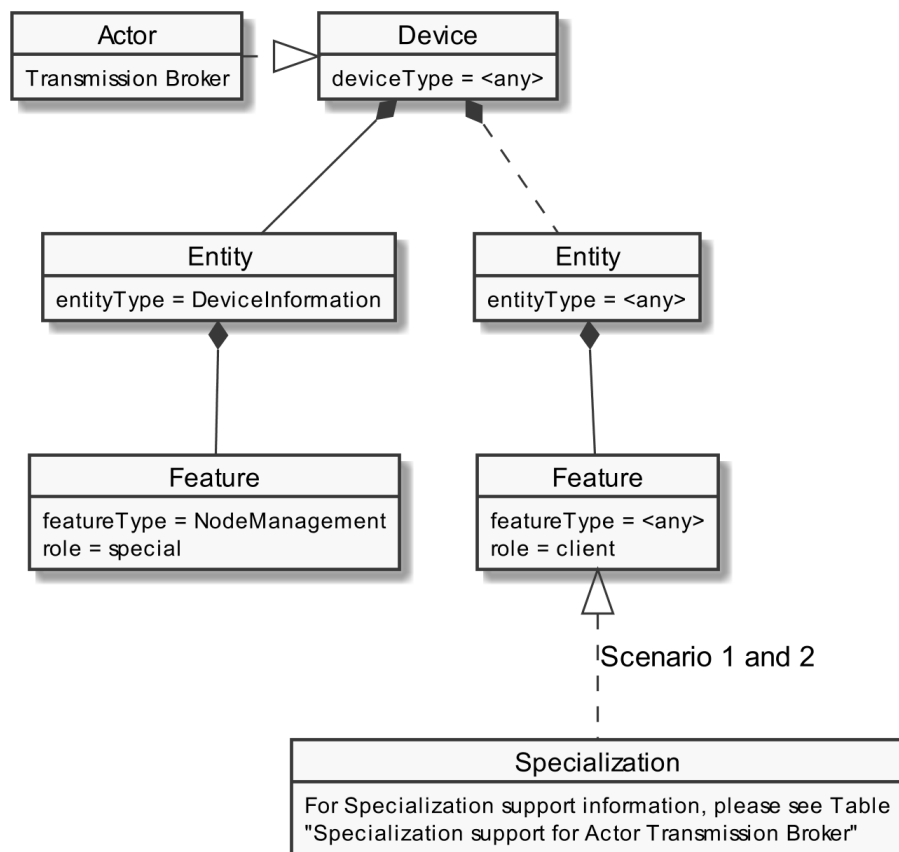


Figure 6: Actor "Transmission Broker" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
TimeSeries_ActivePowerForecast	1	Table 13
ElectricalConnection_EntityContractualConsumptionNominalMax	2	Table 14
ElectricalConnection_EntityContractualProductionNominalMax	2	Table 15

Table 12: Specialization support for Actor Transmission Broker

3.2.2.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.2.3 Client data - Specializations

3.2.2.3.1 Topic "TimeSeries"

3.2.2.3.1.1 Specialization "TimeSeries_ActivePowerForecast"

Scenario {...}: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
1: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
1: O	measurementId	<m*(1..1)>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
1: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
1: M	scopeType	"activePowerForecast"	
1: M	TimeSeries. timeSeriesListData. timeSeriesData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: F	timePeriod.		
1: M	timeSeriesSlot.		[PODF-003]
1: M	timeSeriesSlot. timeSeriesSlotId		
1: M	timeSeriesSlot. timePeriod.		
1: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[PODF-001]
1: M	timeSeriesSlot. duration		[PODF-002], [PODF-007] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The

			duration SHALL last to the start time of the next slot, if any.
1: M	timeSeriesSlot. value.		[PODF-004], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal. See [Scaled number rules].
1: M	timeSeriesSlot. value. number		
1: M	timeSeriesSlot. value. scale		
1: M	timeSeriesSlot. minValue.		[PODF-005], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
1: M	timeSeriesSlot. minValue. number		
1: M	timeSeriesSlot. minValue. scale		
1: M	timeSeriesSlot. maxValue.		[PODF-006], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
1: M	timeSeriesSlot. maxValue. number		
1: M	timeSeriesSlot. maxValue. scale		

Table 13: Content of Specialization "TimeSeries_ActivePowerForecast" at Actor Transmission Broker

3.2.2.3.2 Topic "ElectricalConnection"

3.2.2.3.2.1 Specialization "ElectricalConnection_EntityContractualConsumptionNominalMax"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#{1..1}>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicId	<cc1#{1..1}>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicContext	"entity"	
2: M	characteristicType	"contractualConsumptionNominalMax"	
2: M	value.		[PODF-009], [PODF-021] [Scaled number rules]
2: M	value. number	≥0	SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	unit	"W"	

Table 14: Content of Specialization "ElectricalConnection_EntityContractualConsumptionNominalMax" at Actor
Transmission Broker

3.2.2.3.2.2 Specialization "ElectricalConnection_EntityContractualProductionNominalMax"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#{1..1}>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicId	<cc2#{1..1}>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicContext	"entity"	
2: M	characteristicType	"contractualProductionNominalMax"	
2: M	value.		[PODF-009], [PODF-022] [Scaled number rules]

2: M	value. number	≥0	SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	unit	"W"	

Table 15: Content of Specialization "ElectricalConnection_EntityContractualProductionNominalMax" at Actor Transmission Broker

3.2.3 CEM

3.2.3.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "CEM" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor CEM's resource hierarchy.

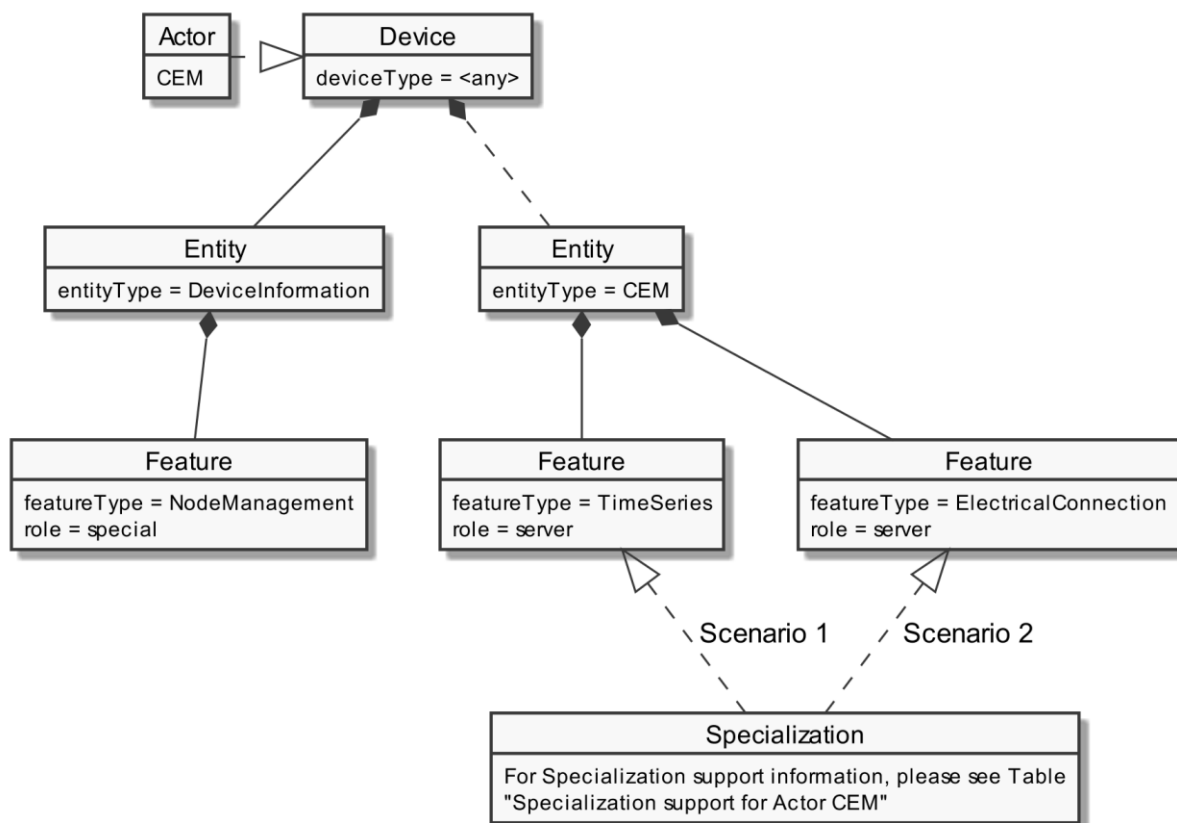


Figure 7: Actor "CEM" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind the entityType "CEM". The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
TimeSeries_ActivePowerForecast	1	TimeSeries	Table 18 Table 19
ElectricalConnection_EntityContractualConsumptionNominalMax	2	ElectricalConnection	Table 20
ElectricalConnection_EntityContractualProductionNominalMax	2	ElectricalConnection	Table 20

Table 16: Specialization support for Actor CEM

3.2.3.2 Server data - Resources

3.2.3.2.1 Overview

Behind the entityType "CEM", the Actor CEM SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
TimeSeries	1: M	timeSeriesDescriptionListData	read (M). partial (R)
	1: M	timeSeriesListData	read (M). partial (R)
ElectricalConnection	2: M	electricalConnectionCharacteristicListData	read (M). partial (R)

Table 17: Feature Types and Functions used within this Use Case by the Actor CEM

For each of these Feature Types the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the CEM acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

3.2.3.2.2 Feature Type "TimeSeries"

3.2.3.2.2.1 Function "timeSeriesDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	timeSeriesType	"plan"	"plan" SHALL be used to model how a certain value will change over time.
1: M	timeSeriesWriteable	"false"	If set to false or omitted, this timeSeriesData SHALL NOT be writeable.
1: O	measurementId	<m*#(1..1)>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
1: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
1: M	scopeType	"activePowerForecast"	

Table 18: Content of Function "timeSeriesDescriptionListData" at Actor CEM

965 3.2.3.2.2.2 Function "timeSeriesListData"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries.timeSeriesListData. timeSeriesData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: F	timePeriod.		
1: M	timeSeriesSlot.		[PODF-003]
1: M	timeSeriesSlot. timeSeriesSlotId		
1: M	timeSeriesSlot. timePeriod.		
1: M	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[PODF-001]
1: M	timeSeriesSlot. duration		[PODF-002], [PODF-007] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
1: M	timeSeriesSlot. value.		[PODF-004], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater or equal. If maxValue is set the content of value SHALL be smaller or equal. See [Scaled number rules].
1: M	timeSeriesSlot. value. number		
1: O	timeSeriesSlot. value. scale		
1: O	timeSeriesSlot. minValue.		[PODF-005], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of minValue SHALL be smaller or equal. If value is not set but maxValue is set the content of minValue SHALL be smaller or equal. See [Scaled number rules].
1: M	timeSeriesSlot. minValue. number		

1: O	timeSeriesSlot. minValue. scale		
1: O	timeSeriesSlot. maxValue.		[PODF-006], [PODF-008], [PODF-011] A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value is set the content of maxValue SHALL be greater or equal. If value is not set but minValue is set the content of maxValue SHALL be greater or equal. See [Scaled number rules].
1: M	timeSeriesSlot. maxValue. number		
1: O	timeSeriesSlot. maxValue. scale		

Table 19: Content of Function "timeSeriesListData" at Actor CEM

3.2.3.2.3 Feature Type "ElectricalConnection"

3.2.3.2.3.1 Function "electricalConnectionCharacteristicListData"

Scenario {...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
2: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicId	<cc1#(1..1)>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicContext	"entity"	
2: M	characteristicType	"contractualConsumptionNominalMax"	
2: M	value.		[PODF-009], [PODF-021] [Scaled number rules]
2: M	value. number	≥0	SHALL be used.
2: M	value. scale		SHALL be used.
2: M	unit	"W"	
2: M	ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData.		
2: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p1#(1..1)>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicId	<cc2#(1..1)>	SHALL be set as SUB IDENTIFIER.
2: M	characteristicContext	"entity"	

2: M	characteristicType	"contractualProductionNominalMax"	
2: M	value.		[PODF-009], [PODF-022] [Scaled number rules]
2: M	value. number	≥0	SHALL be used.
2: M	value. scale		SHALL be used.
2: M	unit	"W"	

Table 20: Content of Function "electricalConnectionCharacteristicListData" at Actor CEM

3.2.3.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

- a) Detailed discovery: allows to discover resource addresses.
- b) Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
- c) Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up to date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

1002 Often the necessary messages of different Scenarios can be combined, so that only one single
1003 message is needed instead of multiple messages for the different Scenarios. This also is the case for
1004 the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery
1005 is necessary, as well as only one subscription request or binding request is needed for each Feature.
1006 Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or
1007 binding is necessary.

1008

1009 **3.3.2 Detailed discovery**

1010 For the functionality where a client already has current detailed discovery information (i.e.
1011 independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

1012 Otherwise, the following procedure SHALL be performed in the given order:

- 1013 1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL
1014 acquire a subscription according to the figure provided within this sub-section.
- 1015 2. A client SHALL read the detailed discovery information according to the figure provided
1016 within this sub-section. It SHALL keep the received information as far as it concerns
1017 mandatory and supported optional Entity Types, Feature Types and Functions of this Use
1018 Case that are needed by the client. This means that a client may choose how to store the
1019 necessary information. E.g. a client Actor can store the information how to address the
1020 necessary Features of the implemented Scenarios but may discard the Entity information.
- 1021 3. If and as long as a client has a subscription to the detailed discovery information of an Actor
1022 and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed
1023 discovery information) the received information as far as it concerns mandatory and
1024 supported optional Entity Types, Feature Types and Functions of this Use Case.

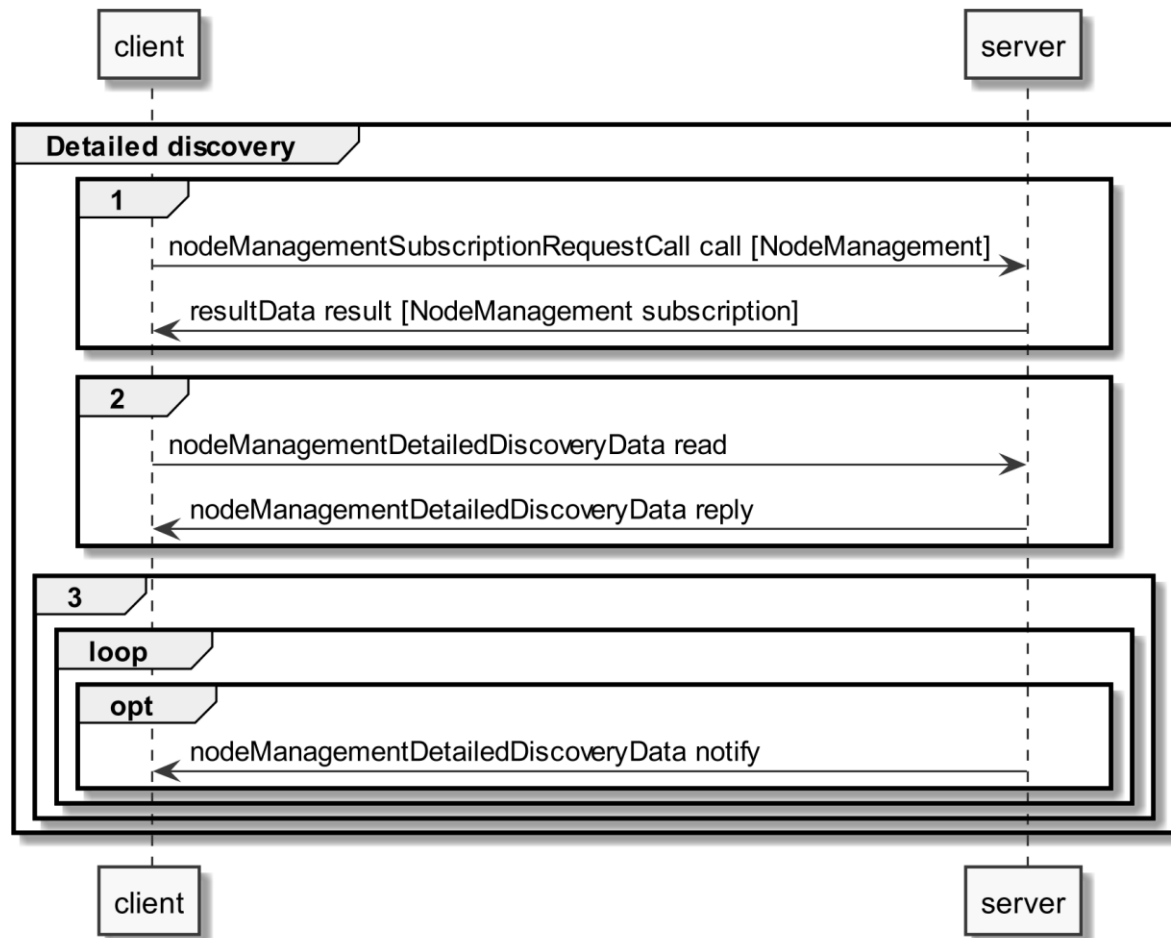


Figure 8: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

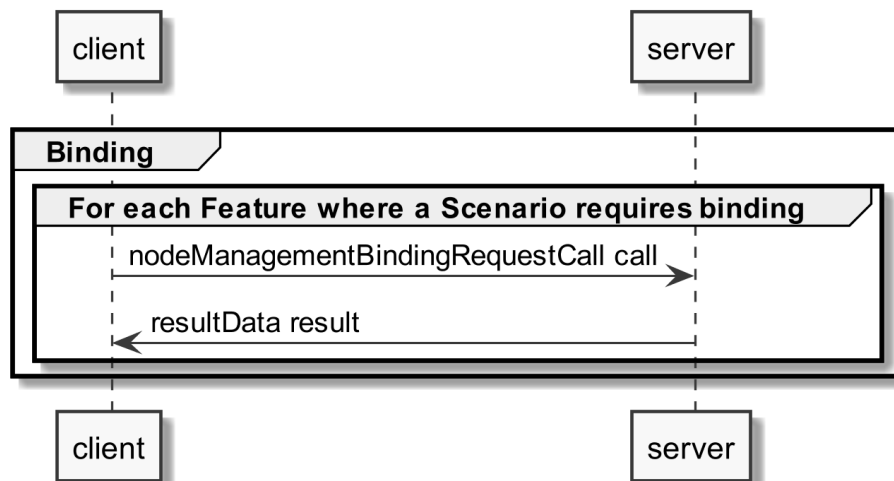


Figure 9: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

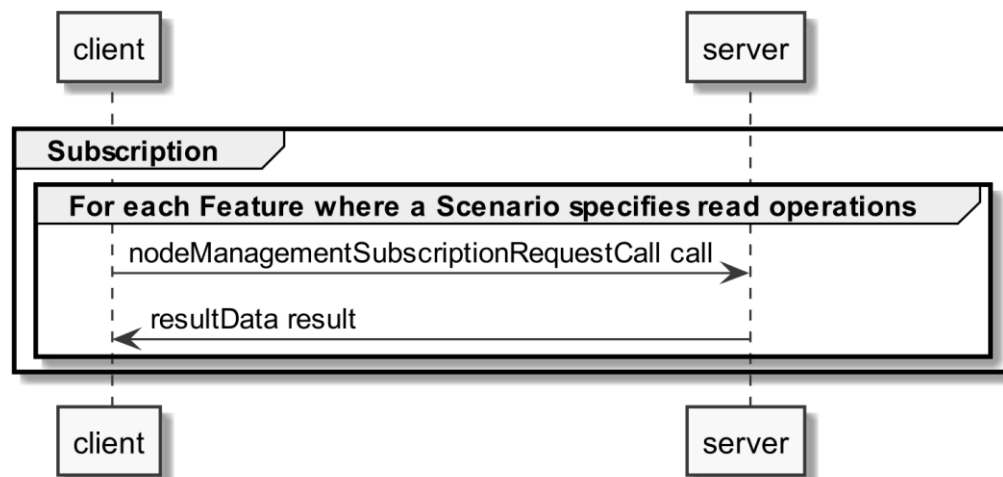


Figure 10: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - CEM's active power forecast

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Binding SHOULD NOT be used for this Scenario.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as

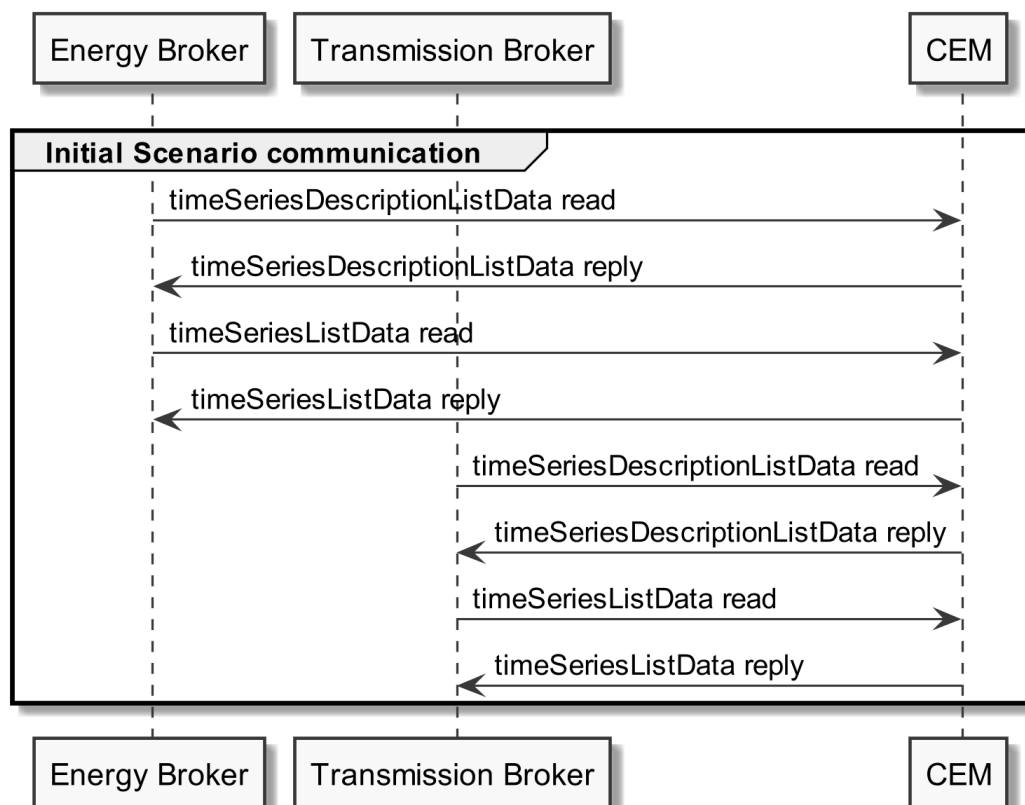
1095 soon as the address of a required resource is known, the Initial Scenario communication for this
 1096 resource MAY start already, even if the addresses of other required resources are not known yet.

1097 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1098 communication is triggered again for those resources.

1099

1100 **3.4.1.2 Initial Scenario communication**

1101 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1102 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1103 resources may have changed in the meantime:



1104

1105 *Figure 11: Scenario 1 - Initial Scenario communication sequence diagram*

1106 **Partial data information regarding [PODF-011]:**

1107 The `timeSeriesDescriptionListData read` SHOULD be a "partial" read operation with the following
 1108 Selector:

- 1109 - `timeSeriesType = "plan"`
- 1110 - `scopeType = "activePowerForecast"`

1111 The `timeSeriesListData read` SHOULD be "partial" read operation with the following Selector:

- 1112 - `timeSeriesId` (derived from the `timeSeriesDescriptionListData reply`)

1113 Note: If partial read is not supported, a full read SHALL be performed.

1114 The following table shows where the required content of the messages from the sequence diagram is
 1115 described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 18	1
timeSeriesListData reply	Table 19	1

1116 *Table 21: Initial Scenario communication content references for Scenario 1*

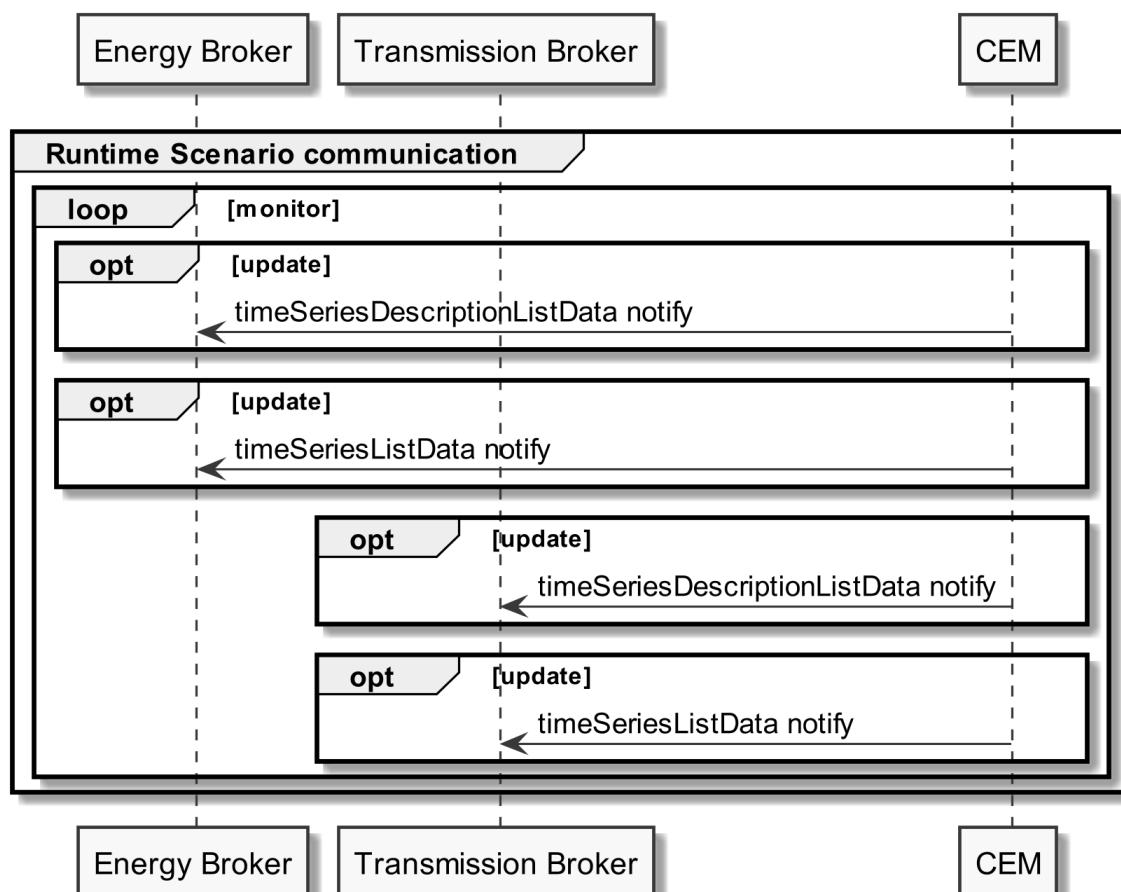
1117 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1118 provided completely, but later during Runtime Scenario communication.

1119

1120 3.4.1.3 Runtime Scenario communication

1121 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1122 during runtime.

1123 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1124 in the following figure:



1125

1126 *Figure 12: Scenario 1 - Runtime Scenario communication sequence diagram*

1127 Note: Normally, in this Scenario only the "timeSeriesListData" Function changes during runtime.
 1128 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1129 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
1130 Scenario.

1131 For timeSeriesDescriptionListData notify and timeSeriesListData notify "partial" delete notifications
1132 SHOULD be supported with the Selector:

1133 - timeSeriesId

1134 Note: To interpret partial notification messages correctly, the information obtained during the Initial
1135 Scenario communication phase is required.

1136 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
1137 not be evaluated.

1138

1139 The following table shows where the required content of the messages of the sequence diagram is
1140 described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData notify	Table 18	1
timeSeriesListData notify	Table 19	1

1141 *Table 22: Runtime Scenario communication content references for Scenario 1*

1142

1143 **3.4.1.4 Additional information**

1144 None.

1145

1146 **3.4.2 Scenario 2 - Constraints**

1147 **3.4.2.1 Pre-Scenario communication**

- 1148 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
1149 of the server Features used in the Initial Scenario communication. If the address of a
1150 particular server Feature is not known, the detailed discovery must be used, as described in
1151 section 3.3.2.
- 1152 2. **Binding:** Binding SHOULD NOT be used for this Scenario.
- 1153 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
1154 the corresponding Actor within this Scenario, as described in section 3.3.4.

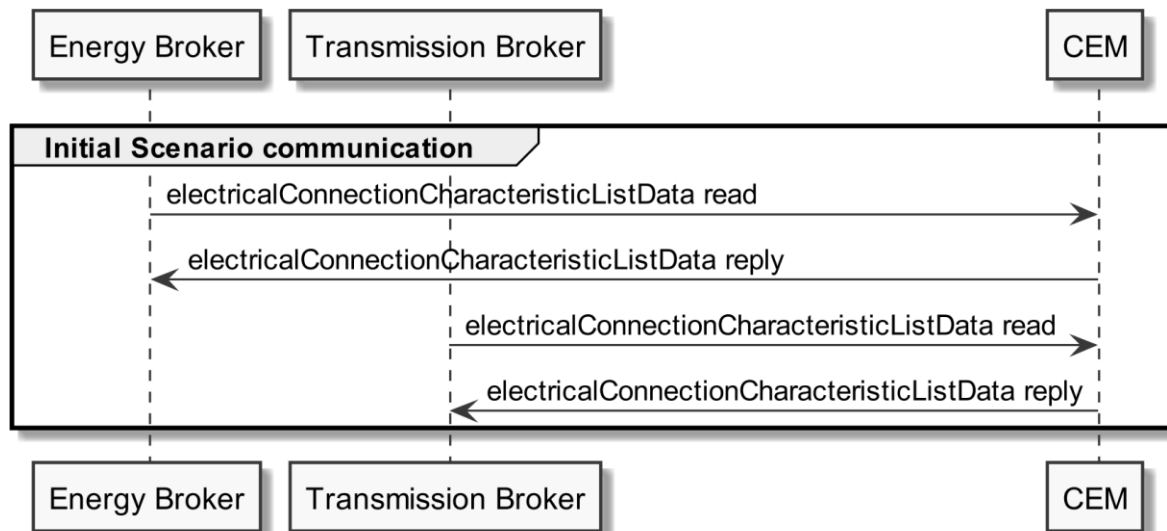
1155 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
1156 are known and the necessary binding and subscription procedures have been finished. However, as
1157 soon as the address of a required resource is known, the Initial Scenario communication for this
1158 resource MAY start already, even if the addresses of other required resources are not known yet.

1159 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
1160 communication is triggered again for those resources.

1161

1162 **3.4.2.2 Initial Scenario communication**

1163 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1164 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1165 resources may have changed in the meantime:



1166

1167 *Figure 13: Scenario 2 - Initial Scenario communication sequence diagram*1168 **Partial data information regarding [PODF-021]:**

1169 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
 1170 following Selectors:

- 1171 - characteristicContext = "entity"
- 1172 - characteristicType = "contractualConsumptionNominalMax"

1173 Note: If partial read is not supported a full read SHALL be performed.

1174

1175 **Partial data information regarding [PODF-022]:**

1176 The electricalConnectionCharacteristicListData read SHOULD be a "partial" read operation with the
 1177 following Selectors:

- 1178 - characteristicContext = "entity"
- 1179 - characteristicType = "contractualProductionNominalMax"

1180 Note: If partial read is not supported a full read SHALL be performed.

1181

1182 The following table shows where the required content of the messages from the sequence diagram is
 1183 described:

Message name from sequence diagram	Content description in table	Scenario number in table
electricalConnectionCharacteristicListData reply	Table 20	2

Table 23: Initial Scenario communication content references for Scenario 2

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.2.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

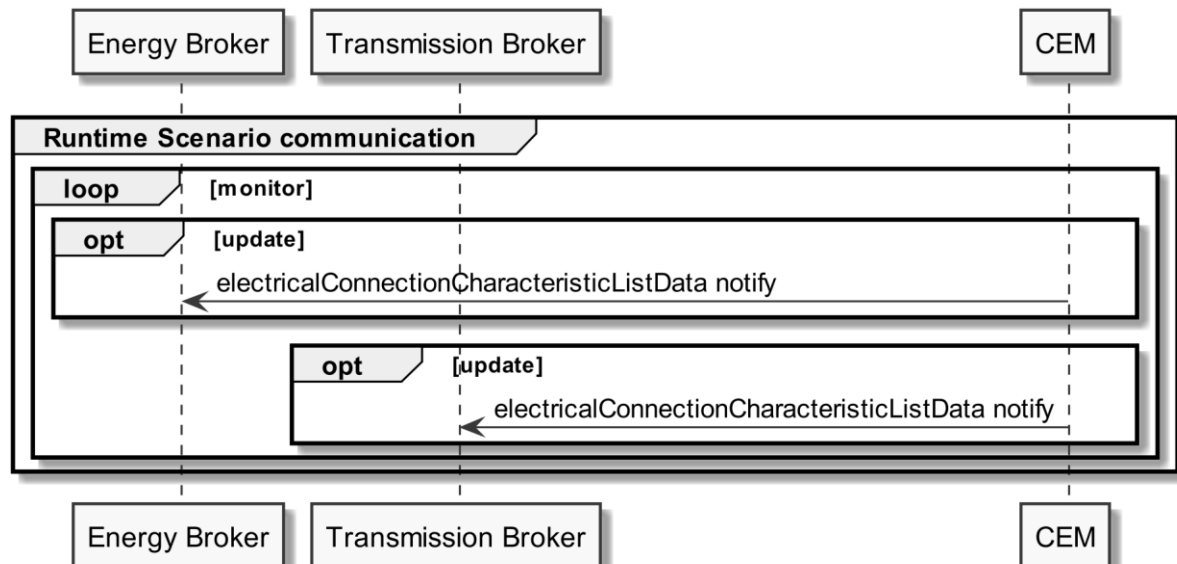


Figure 14: Scenario 2 - Runtime Scenario communication sequence diagram

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For electricalConnectionCharacteristicListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- characteristicId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

1207 The following table shows where the required content of the messages of the sequence diagram is
1208 described:

Message name from sequence diagram	Content description in table	Scenario number in table
electricalConnectionCharacteristicListData notify	Table 20	2

1209 *Table 24: Runtime Scenario communication content references for Scenario 2*

1210

1211 **3.4.2.4 Additional information**

1212 None.

1213